

Kadir Kadiroğlu

Mechatronics Engineer

[kdroglu.github.io](#) [+90 544 242 56 09](#) [kadir.k1998@gmail.com](#) [LinkedIn Profile](#) [İstanbul, Türkiye](#)

PROFESSIONAL SUMMARY

Mechatronics Engineer specializing in embedded systems, mechanical design, and control theory. Demonstrated expertise in integrating mechanical, electrical, and software components through complex projects including haptic feedback robotic systems and autonomous ROVs. Proficient in Fusion 360, C++, and PCB design, delivering robust engineering solutions through hands-on innovation and technical precision.

TECHNICAL SKILLS

MECHANICAL DESIGN & CAD

Fusion 360 (Generative Design, Simulation, Animation) SolidWorks
3D Printing Rapid Prototyping

PROGRAMMING

C++ (Embedded) Java (OOP & GUI) C# (Windows Forms)
MATLAB & Simulink Python

AUTOMATION

PLC Programming (Panasonic FPWIN Pro)
HMI Design (EasyBuilder)

ELECTRONICS & PCB

KiCad (PCB Design) Circuit Design
Sensors & Actuators Hardware Integration

EMBEDDED SYSTEMS

Arduino ESP32 Raspberry Pi Pico STM32

LANGUAGES

English (B2 Proficiency) Turkish (Native)

PROFESSIONAL EXPERIENCE

Mechatronics Engineering Intern

26 Feb 2024 - 22 Mar 2024

Lenta Marine İnsansız Su Altı Sistemleri Ltd.Şti

- Spearheaded the design and integration of ROV electronic systems using Pixhawk 2.4.8 flight controllers and Raspberry Pi 3B for autonomous underwater navigation.
- Developed a Python (PyQt5) Operator Interface with MAVLink communication and OpenCV for real-time roll/pitch visualization.
- Configured BLDC thrusters with bidirectional ESCs (BLHeliSuite) and assembled watertight housings with 4S Li-ion battery systems.

R&D Engineering Intern

28 Aug 2023 - 25 Sep 2023

Alarge Alıç Ağacı Makine Elektronik Arge Ltd.Şti

- Executed R&D for industrial testing devices (Vicat, Vacuum Leak, MFI, and Hydrostatic Pressure) by programming Panasonic PLCs (FPWIN Pro) and designing HMI screens (EasyBuilder).
- Developed a closed-loop PID control algorithm for DC motors using Arduino and magnetic encoders for an Automatic Film Application Device.
- Assembled and tested electronic panels, pneumatic circuits, and hydraulic systems for high-pressure testing equipment.

Workshop Intern

30 Jan 2023 - 24 Feb 2023

Yıldız Teknik Üniversitesi Teknopark Prototip Atölyesi

- Manufactured precision parts, such as hex nipples, using manual and CNC machining operations including Turning, Milling, Drilling, and Tapping.
- Generated G-codes using SolidCAM and executed operations on CNC Lathe and Milling machines.
- Managed workshop inventory, performed machinery maintenance, and applied strict safety protocols for industrial equipment.

KEY PROJECTS & RESEARCH

Graduation Project: Haptic Feedback Robotic Arm System

C++, Fusion 360, Control Theory, Sensors

- Engineered a Master-Slave robotic system featuring haptic feedback algorithms, kinematic coupling, and real-time control loops.
- Implemented bidirectional communication protocols allowing the operator to physically feel forces exerted on the slave arm.
- Designed and fabricated mechanical components using Fusion 360 and rapid prototyping techniques to ensure structural integrity.
- Optimized control algorithms to minimize latency and enhance the fidelity of the force feedback mechanism.

Autonomous Pan/Tilt Camera Tracking System

ESP32-CAM, Servo Motors, HTML/Web, Computer Vision

- Developed a dual-axis camera system capable of autonomous face and object tracking using ESP32-CAM and servo motors.
- Created a responsive HTML/Web user interface for low-latency video streaming and manual joystick control via mobile devices.

Realistic RC Car Simulator

Embedded Systems, Physics Simulation, Mechanical Design

- Built a custom RC car controller with a steering wheel and pedals to simulate realistic driving dynamics.
- Implemented advanced algorithms to simulate vehicle inertia, acceleration, and deceleration physics for an immersive experience.

Magnetic Levitation Control System

PID Control, Hall Effect Sensors, Electromagnets, 3D Printing

- Designed a PID-controlled magnetic levitation system using Hall effect sensors and electromagnets to suspend objects in mid-air.
- Created a 3D-printed aesthetic frame and rotating mechanism to enhance the visual illusion and system stability.

High-Efficiency Planetary Gearbox Design

Fusion 360, Mechanical Engineering, 3D Printing

- Designed a 25:1 reduction ratio planetary gearbox for BLDC motors to maximize torque while minimizing volume.
- Calculated gear dimensions, bearing selection, and bolt placement in Fusion 360, and fabricated all components via 3D printing.

EDUCATION

Yıldız Technical University

Bachelor of Science in Mechatronics Engineering (GPA: 2.88)

Graduated: June 2024